

1 Log-to-Vec Benchmark — Comprehensive Research Report

Generated: 2026-05-05

1.1 1. Project Goal

Learn useful **embeddings of PLC/SCADA log files** for mode-change detection and regime identification in industrial control systems. The logs are timestamped records of sensor readings and discrete machine state codes — essentially serialised multivariate time series with no meaningful free text.

The motivating question: can a learned representation capture operating regime, load, and mode-change structure **without hand-labelled data**, well enough to be useful downstream?

1.2 2. Experiment History at a Glance

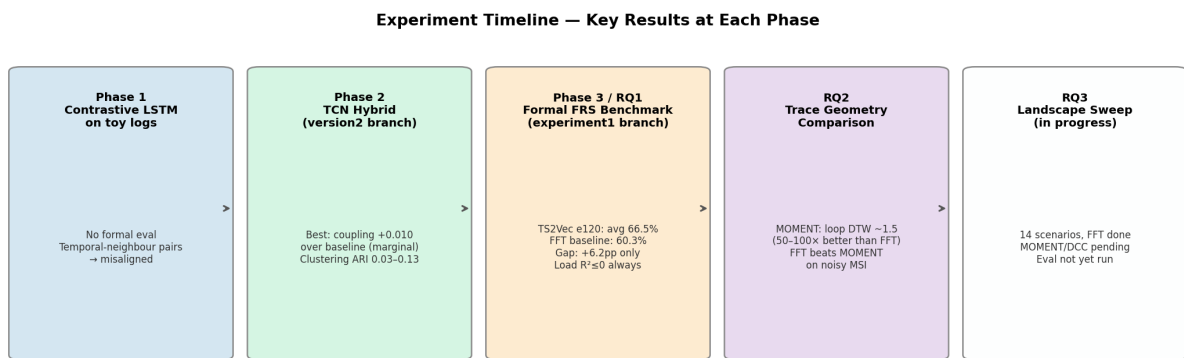


Figure 1: Phase timeline

1.3 3. Phase 1 — Contrastive LSTM on Toy Logs

Setup

- LSTM encoder trained with NT-Xent contrastive loss.
- Positive pairs: temporal neighbours within a trajectory.
- Data: toy CSV logs generated synthetically (`data/toy_logs.csv`); sine-wave generator examples.
- Evaluation: ad-hoc clustering and retrieval visualisations only. No formal metric suite.

Outcome No quantitative results recorded. Discontinued because:

1. Temporal-neighbour positives teach "what is nearby in time", not "what shares a mode".
2. Toy logs had no ground-truth factorised labels — claims about quality were unverifiable.

Lesson: A controlled synthetic benchmark with known latent structure is a prerequisite before any learning claim can be made.

1.4 4. Phase 2 — TCN Hybrid on FSSS (version2 branch)

Setup

- TCN (Temporal Convolutional Network) with dilated residual conv blocks.

- Hybrid loss: NT-Xent contrastive + reconstruction (masked span, Gaussian noise, time shift).
- Data: FSSS datasets — earlier, smaller synthetic variants of FRS.
- Formal eval suite added: linear and RBF probes, retrieval@K, clustering ARI.
- Baselines: raw statistics, FFT, PCA.

Results (RBF probe balanced accuracy)

Factor	Baseline (best)	Learned (best)	Delta
mode_id	0.277	0.267	−0.010 — baseline wins
spectral_id	0.701	0.653	−0.048 — baseline wins
coupling_id	0.413	0.423	+0.010 — marginal
transition	0.558	0.534	−0.024 — baseline wins
mean_load	≈0	≈0	tied at zero

Retrieval R@5 was misleadingly high (0.961–0.993) due to temporal-neighbour training bias. Clustering ARI was very poor (0.020–0.134).

Lesson: TCN + temporal-neighbour contrastive + reconstruction does not extract factorised structure. The objective is misaligned with the evaluation goal. Baselines are formidable.

1.5 5. Phase 3 / RQ1 — Formal FRS Benchmark

1.5.1 5.1 Dataset

FRS (Factorised Regime Sequence) — a controlled synthetic benchmark with known latent structure.

Each trajectory is governed by three independent latent factors:

- **Mode ID** — discrete operating regime (sine-wave pattern per channel)
- **Spectral ID** — frequency configuration
- **Coupling ID** — cross-channel interaction strength
- **Load** — continuous amplitude modulation (hardest)

Two datasets: `frs_clean_vnext_long` ($\sigma_{\text{noise}} = 0.05$) and `frs_noisy_vnext_long` ($\sigma_{\text{noise}} = 0.20$). Both use a **trajectory-level 70/15/15 split** to prevent leakage.

1.5.2 5.2 Methods

Method	Description	Seeds
MOMENT (pretrained)	Foundation model, 768-dim embeddings, frozen	3 (42/53/64)
TS2Vec-style e80	Trained from scratch, 64-dim, 80 epochs	1
TS2Vec-style e120	Same, tuned to 120 epochs	1
FFT baseline	Per-channel amplitude spectrum, 14–56 dim	1
Summary baseline	Mean, std, min, max, per channel	1

1.5.3 5.3 Evaluation Suite

- **Probes:** linear and RBF SVM probes for each latent factor → balanced accuracy

- **Retrieval:** precision@10 for each factor
- **Clustering:** K-Means ARI for each factor
- **Transition:** detection of mode-change windows

1.5.4 5.4 Results

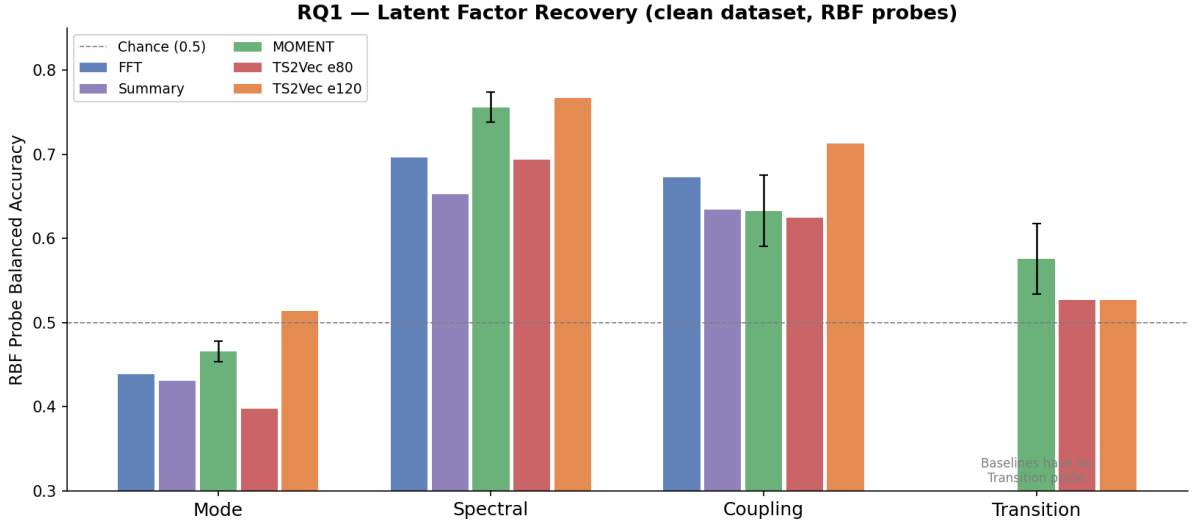


Figure 2: RQ1 probe accuracy

Table 1 — RBF Probe Balanced Accuracy (clean dataset)

Method	Mode	Spectral	Coupling	Transition	Load R^2
FFT baseline	0.439	0.696	0.673	—	−0.006
Summary baseline	0.431	0.653	0.634	—	−0.006
MOMENT	0.466 ± 0.012	0.756 ± 0.018	0.633 ± 0.042	0.576 ± 0.042	-0.037 ± 0.049
TS2Vec e80	0.398	0.694	0.625	0.527	−0.006
TS2Vec e120	0.514	0.767	0.713	0.527	−0.006

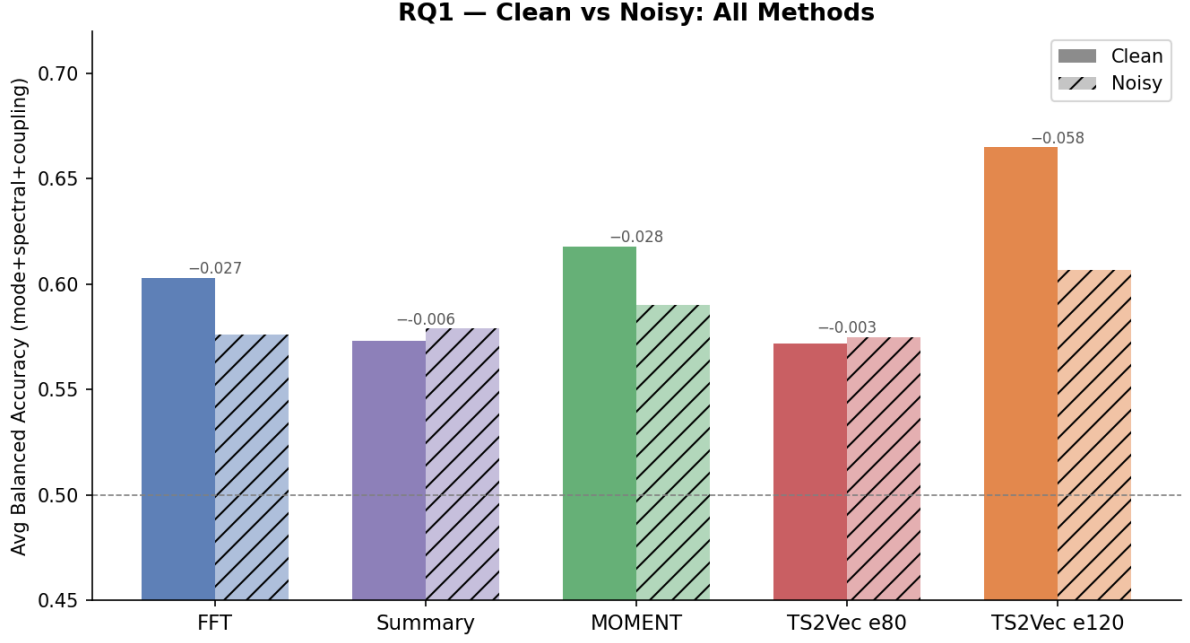


Figure 3: Clean vs noisy comparison

Table 2 — Average balanced accuracy (mode + spectral + coupling)

Method	Clean	Noisy	Drop
FFT baseline	0.603	0.576	−0.027
Summary baseline	0.573	0.579	+0.006
MOMENT	0.618	0.590	−0.028
TS2Vec e80	0.572	0.575	+0.003
TS2Vec e120	0.665	0.607	−0.058

The margin of the best learned method over the best baseline is only **+6.2 pp** on clean data, narrowing to **+3.1 pp** on noisy data.

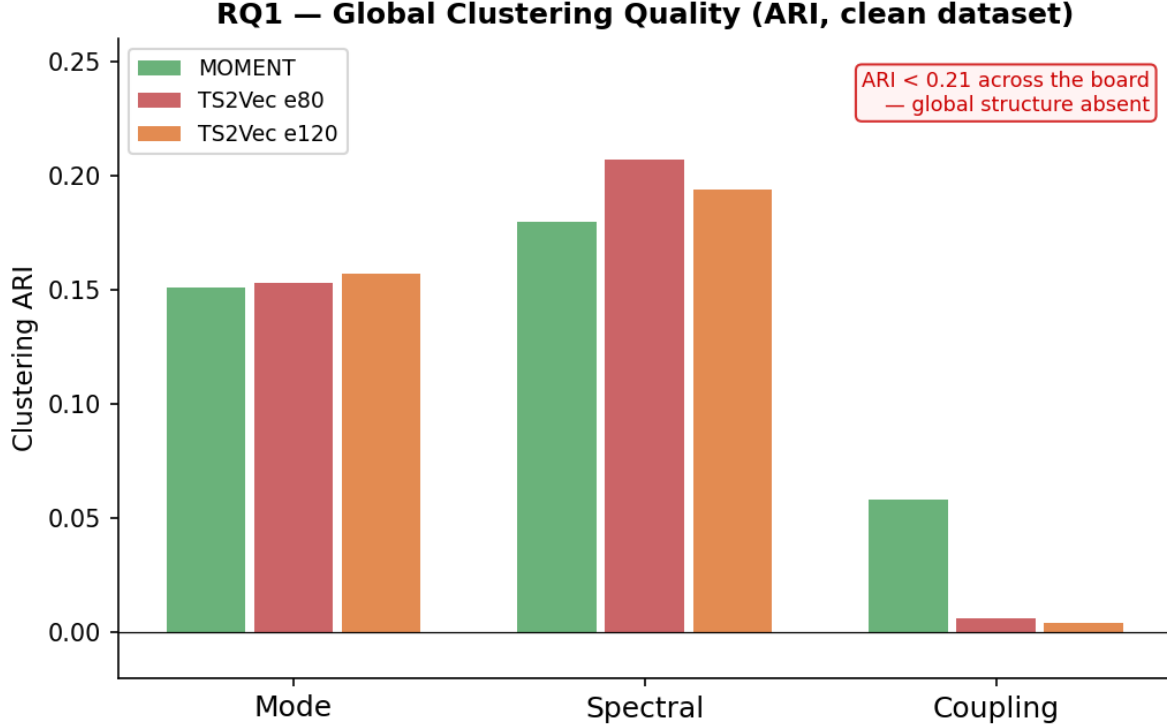


Figure 4: Clustering ARI

Table 3 — Clustering ARI (K-Means, clean dataset)

Method	Mode	Spectral	Coupling	Avg
MOMENT	0.151	0.180	0.058	0.130
TS2Vec e80	0.153	0.207	0.006	0.122
TS2Vec e120	0.157	0.194	0.004	0.118

Global clustering quality is uniformly poor ($\text{ARI} < 0.21$). More epochs (e80→e120) helps probes but **degrades** coupling clustering.

Critical finding: Load is completely unlearned. Load $R^2 \leq 0$ for every method in every setting.

1.6 6. RQ2 — Trace Geometry Comparison

1.6.1 6.1 Setup

Three synthetic problems modelled after CNC-machining analogies:

Problem	Description	Channels	Noise
P1 — Simple 1D	Clearly separated frequencies	1	low
P2 — Multi-channel	Cross-channel frequency mixing	4	low
P3 — Hard Noisy	Similar frequencies, high noise	4	$\sigma=0.20$

Same three models evaluated: FFT, MOMENT, TS2Vec.

Key question: Does the embedding produce geometrically coherent, mode-separable traces that form closed loops in PCA space?

1.6.2 6.2 Metrics

Metric	Symbol	Direction	Meaning
Mode Separability Index	MSI	↑ better	Inter-mode distance / intra-mode spread
Loop Consistency	Loop DTW	↓ better	DTW distance between repeated mode traces
Transition Sharpness	Trans. Sharp	↓ better	Windows to cross midpoint after a mode change
PCA Compactness	PCA Compact	↓ better	Convex hull area of mode loop in PC1–PC2
Centroid Stability	Centroid Stab	↓ better	Std of per-run mode centroids

1.6.3 6.3 Results

Full metric table:

	MSI ↑	Loop DTW ↓	Trans. Sharp ↓	PCA Compact ↓	Centroid Stab ↓
P1 / FFT	11.1	66.6	3.0	0.50	0.289
P1 / MOMENT	71.1	2.76	5.7	0.23	0.002
P1 / TS2Vec	20.0	44.6	3.0	0.37	0.085
P2 / FFT	22.5	146.1	3.3	0.26	0.278
P2 / MOMENT	27.2	1.43	1.67	1.63	0.001
P2 / TS2Vec	4.4	108.2	3.67	48.4	0.509
P3 / FFT	20.2	167.4	4.0	0.33	0.364
P3 / MOMENT	14.3	1.64	1.33	7.97	0.001
P3 / TS2Vec	4.6	107.8	4.0	55.9	0.529

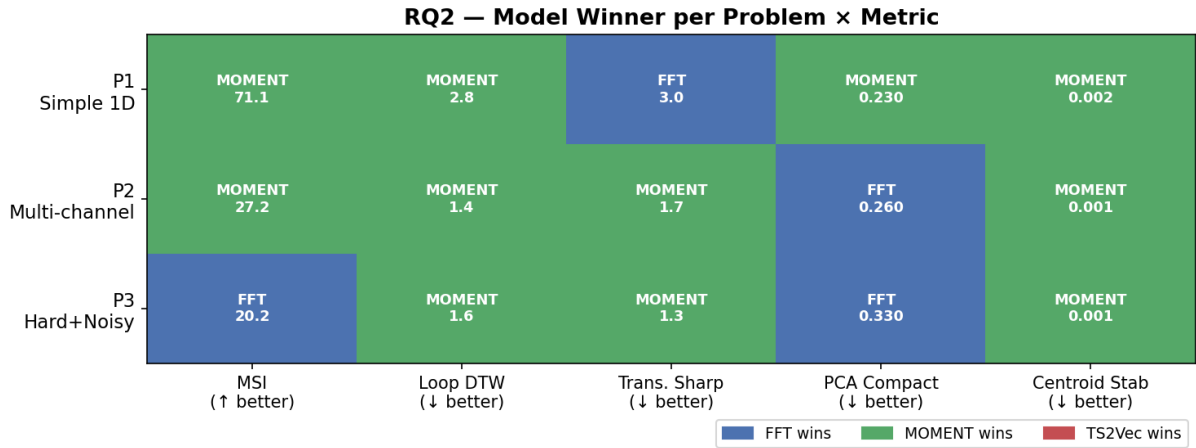


Figure 5: Winner heatmap

RQ2 — Core Embedding Quality Metrics

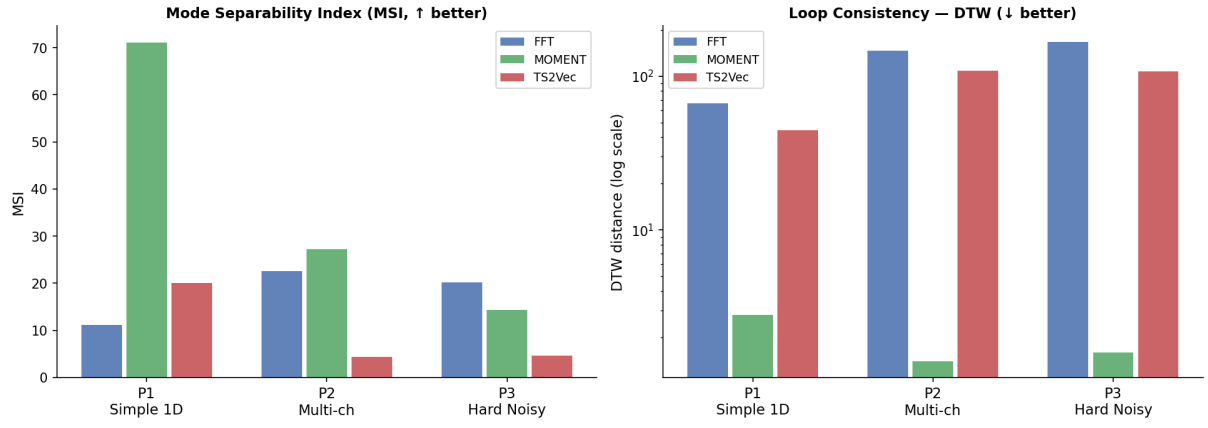


Figure 6: MSI and Loop Consistency

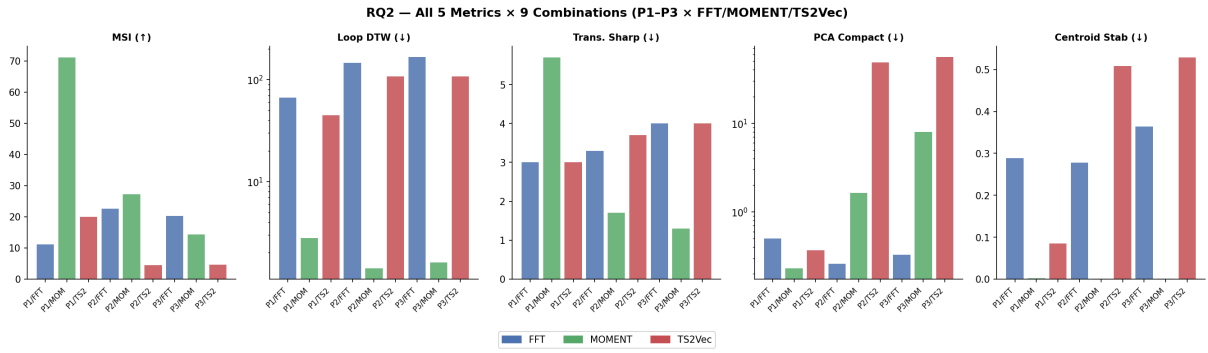


Figure 7: All 5 metrics

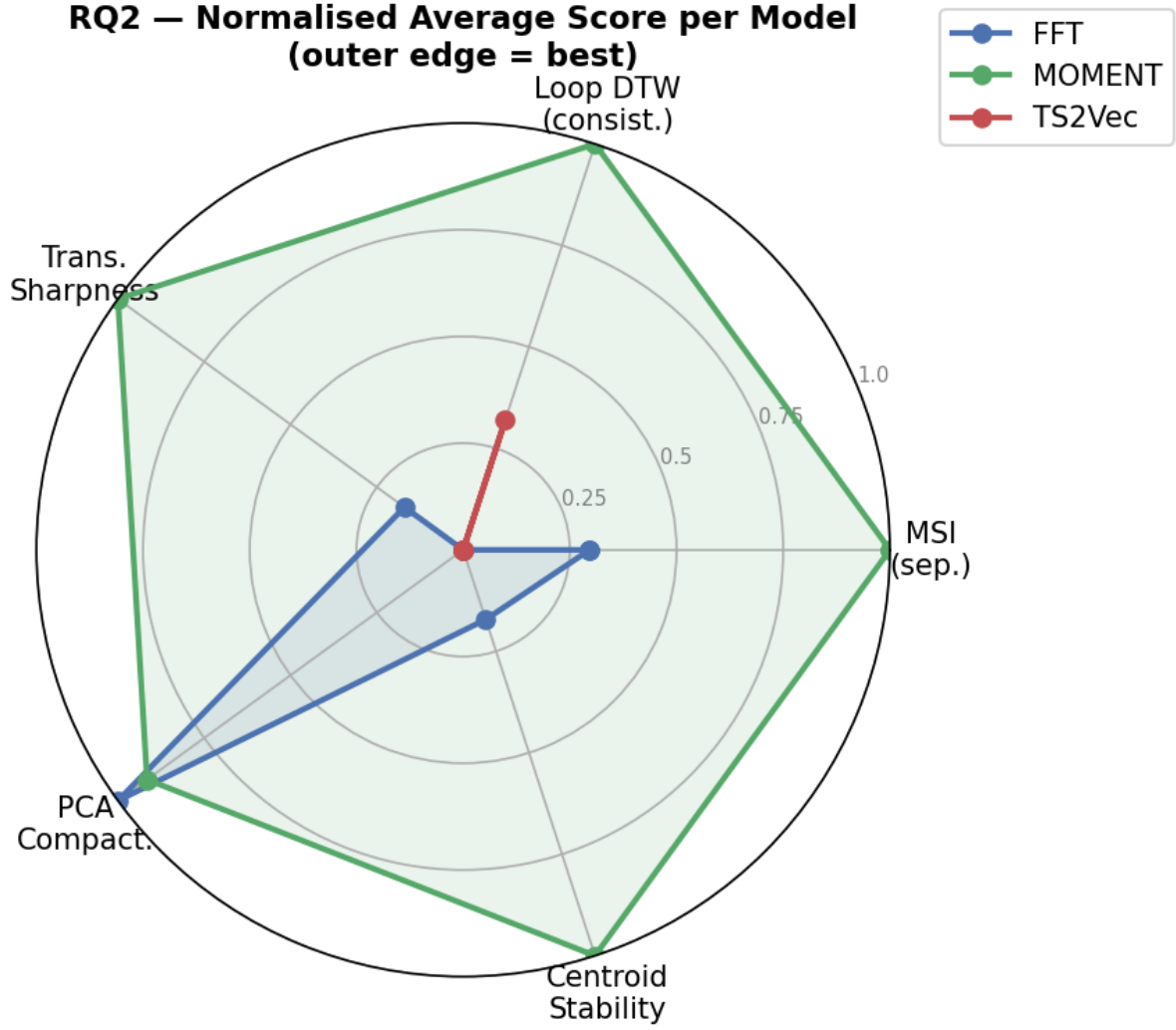


Figure 8: Radar chart

1.6.4 6.4 Key Findings

1. MOMENT’s loop consistency is extraordinary. MOMENT produces loop DTW of 1.4–2.8 across all three problems, vs 66–167 for FFT and 44–108 for TS2Vec — roughly 20–100× better. Mode traces are tight, repeatable closed orbits in PCA space. This is exactly the property needed for online anomaly detection.

2. MOMENT centroid stability is near-zero. Centroid stability 0.001–0.002 vs 0.08–0.53 for others. Mode regions don’t drift between different test trajectories — critical for a reference-based detector.

3. FFT beats MOMENT on MSI in the hard noisy case (P3): 20.2 vs 14.3. This is the key surprise. MOMENT’s pretrained features can’t separate modes when frequencies are close and noise is high, whereas the raw spectrum remains more discriminative. MOMENT still wins on loop consistency and transition sharpness even in P3 — the modes are compact and transitions are fast, but they overlap more in PCA space.

4. TS2Vec is consistently the worst. Low MSI (4.4–20.0), catastrophic PCA compactness on multichannel (48–56 vs FFT’s 0.26–0.33), centroid stability 0.09–0.53. The 64-dim trained representation fails to find useful geometry in these problems.

5. MOMENT’s PCA compactness degrades on multichannel noisy data. P2: 1.63, P3: 7.97 vs FFT’s 0.26, 0.33. Likely a dimensionality projection artefact — projecting 768 dims to 2 PCA dims may scatter loops — rather than a fundamental embedding failure, since loop consistency and centroid stability remain excellent.

1.7 7. Cross-Experiment Synthesis — The Role Reversal

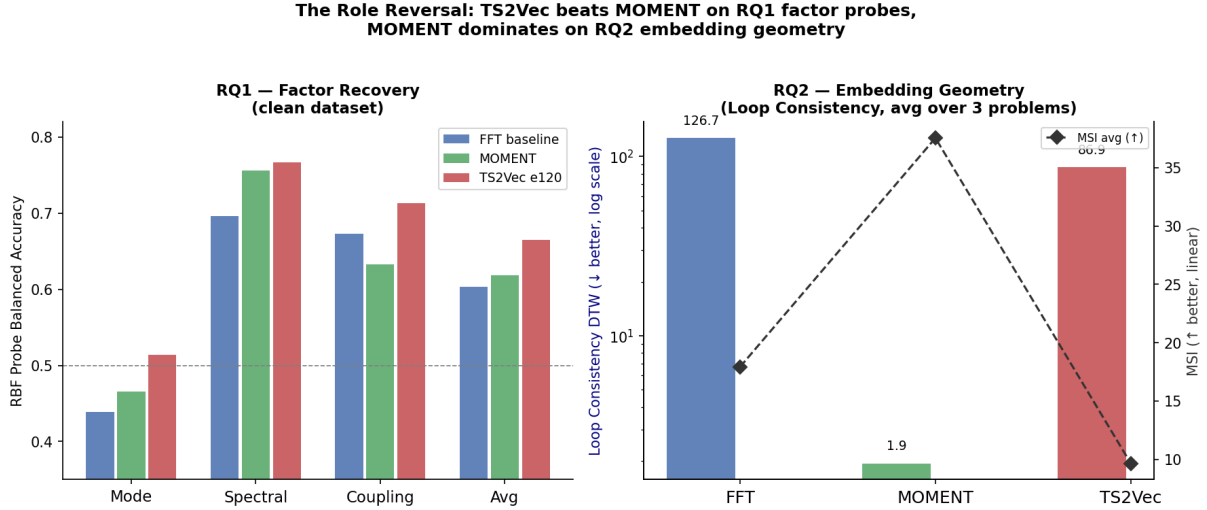


Figure 9: Role reversal

One of the most striking findings is a **role reversal** between MOMENT and TS2Vec across the two experiments:

	RQ1 (factor recovery probes)	RQ2 (embedding geometry)
MOMENT	61.8% avg balanced acc — middle	Loop DTW ~1.5 — dominant
TS2Vec e120	66.5% avg balanced acc — best	Loop DTW ~87 — worst

Interpretation:

- TS2Vec’s contrastive training explicitly pushes same-window representations together and pulls

different windows apart, which helps linear/RBF probes recover discrete factor labels. But the geometry is chaotic — loops don’t form, centroids drift.

- MOMENT’s pretrained features produce geometrically regular embeddings — tight, stable, fast

mode transitions — even without any task-specific training. But its 768-dim space doesn’t encode fine-grained discrete factor boundaries as sharply as TS2Vec’s trained 64-dim space.

These are **complementary strengths**: TS2Vec for discrete label recovery; MOMENT for continuous online monitoring and anomaly detection.

1.8 8. RQ3 — Landscape Sweep (in progress)

1.8.1 8.1 Setup

14 systematically varied scenarios exploring the failure modes of each embedding:

Axis	Levels
Baseline	3-mode, 4-channel, low noise, no missing
Missing data	rand 10%, rand 30%, block 30%, channel missing
Noise	$\sigma=0.00, 0.20, 0.40$
Number of modes	2, 5, 8
Signal type	sine, step, mixed

1.8.2 8.2 Current status

- **Data generation:** complete for all 14 scenarios
- **FFT embeddings (fft8, fft32):** complete for all 14 scenarios
- **MOMENT embeddings:** pending (cluster jobs `rq3_moment.slurm`)
- **DCC embeddings:** pending (cluster jobs `rq3_dcc.slurm`)
- **Metrics, plots, report:** empty — requires all embeddings first

The eval pipeline (`rq3_eval.slurm`) is ready to run on the login node once embeddings land.

1.9 9. Lessons Learned

1.9.1 9.1 Method design

#	Lesson
L1	Temporal-neighbour contrastive pairs teach temporal proximity, not mode identity. Don't use them with
L2	Reconstruction loss adds complexity but doesn't fix misaligned objectives.
L3	More epochs help probes but can harm global structure.
L4	Pretrained foundation models give strong geometry without task-specific training, but may miss fine-gr
L5	Load (continuous amplitude) is not learnable from the current setup — every method gives $R^2 \leq 0$. Fact

1.9.2 9.2 Evaluation design

#	Lesson
L6	Retrieval R@K is misleading when the training objective biases nearest neighbours. Always check cluste
L7	Single-seed results for TS2Vec should be treated cautiously. Variance across seeds can be substantial.
L8	Geometry metrics (loop consistency, centroid stability) reveal embedding quality that probe accuracy m
L9	PCA compactness may penalise high-dimensional embeddings unfairly — a 768→2 projection loses mor

1.9.3 9.3 Benchmark design

#	Lesson
L10	FRS is a valid scientific instrument for the core learning problem, but does not capture named channe
L11	High baseline performance (FFT avg 60.3%) indicates the task has strong spectral structure — learne
L12	Missing data and noise stress-testing (RQ3) is necessary before recommending any method for product

1.10 10. Open Questions

1. Will MOMENT's MSI advantage recover under the RQ3 landscape sweep?
FFT beat MOMENT on P3

MSI — does this generalise to the block-missing or high-noise scenarios in RQ3?

1. **Why does more TS2Vec training degrade coupling clustering while improving probes?** The mode

representations become more linearly separable but globally less organised — suggests the contrastive loss is collapsing intra-mode variance selectively.

1. **Is MOMENT's poor PCA compactness a projection artefact or a real geometry problem?** Would

all-pairs silhouette score in the full 768-dim space tell a different story?

1. **Can factor-aware pairs fix the load problem?** Windows with matching load values as positives

instead of temporal neighbours — the most direct fix for open problem #3 (Load $R^2 \leq 0$).

1. **Does the FFT-vs-MOMENT MSI reversal on noisy data matter for the actual application?**

MOMENT still has faster transitions and more stable centroids even on P3. For online monitoring, loop consistency may matter more than MSI.

1.11 11. Existing Visualisations

RQ2 individual model plots are already available (45 files, 5 plot types $\times$ 9 combinations):

experiments/rq2_trace_comparison/plots/{p1_simple_1d,p2_multichannel,p3_hard_noisy}/{fft,moment,ts2vec}/	
*_worm.png	trajectory trace through PCA
space	
*_mode_loops.png	per-mode closed loops overlaid
*_centroids.png	mode centroids in PCA space
*_distance_heatmap.png	pairwise embedding distances
*_centroid_distance.png	centroid distance over time

Most visually informative contrasts:

- p1_simple_1d_moment_mode_loops.png — best case (MOMENT dominant)
- p3_hard_noisy_fft_mode_loops.png — the surprise (FFT wins MSI on hard problem)
- p2_multichannel_ts2vec_mode_loops.png — worst case (TS2Vec geometry collapse)

1.12 12. Next Steps

1.12.1 Immediate (unblock RQ3)

1. Check whether `rq3_moment.slurm` and `rq3_dcc.slurm` jobs have completed on Alvis.
2. Sync results back, run `rq3_eval.slurm` on login node.

1.12.2 Track A — FRS pipeline fixes (highest priority)

- **A1:** Factor-aware positive pairs for TS2Vec (fix load and coupling learning)
- **A2:** Load regression auxiliary head
- **A4:** Multi-seed TS2Vec to quantify variance
- **A3:** OOD splits (unseen factor combinations at test time)

1.12.3 Track B — PLC log bridge

- **B1:** Synthetic PLC log corpus with FRS-like factors
- **B2:** Tabular/numerical embedding extraction
- **B3:** Cross-modal comparison with FRS eval infrastructure

1.12.4 Track C — Geometry analysis

- Replace PCA compactness with silhouette score in full embedding space
- Investigate MOMENT’s loop consistency advantage: is it universal or data-dependent?